

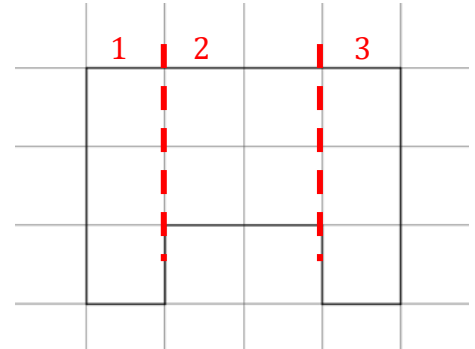
Grade 6 Accelerated
Unit 7
Lesson 6 Homework

Name Answer Key
Date _____
Period _____

A composite figure is shown for questions 1–3. The length of each square is 1 inch (in.).

1. Draw lines to decompose the figure into rectangles.

Answers may vary.



2. Find the area of each rectangle. Include units.

$$\text{Area}_1 = 3 \cdot 1 = 3 \text{ in}^2$$

$$\text{Area}_2 = 2 \cdot 2 = 4 \text{ in}^2$$

$$\text{Area}_3 = 3 \cdot 1 = 3 \text{ in}^2$$

3. Find the sum of the areas of the rectangles to find the area of the composite figure. Include units.

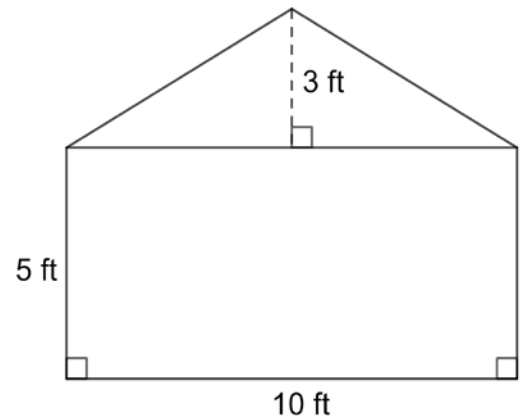
$$3 + 4 + 3 = 10 \text{ in}^2$$

A composite figure measured in feet (ft) is shown for questions 4–5.

4. Find the area of the triangle and the rectangle. Include units.

$$\text{Triangle} \rightarrow 0.5 \cdot 10 \cdot 3 = 15 \text{ ft}^2$$

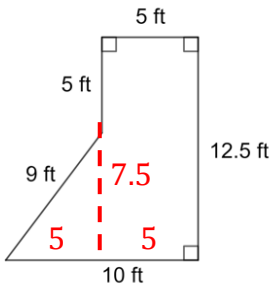
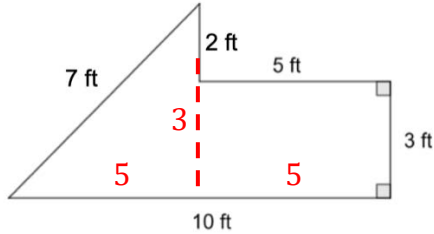
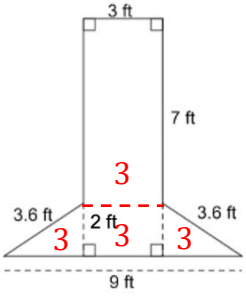
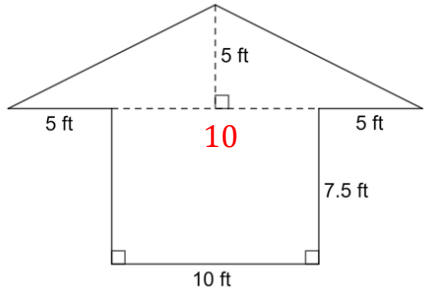
$$\text{Rectangle} \rightarrow 5 \cdot 10 = 50 \text{ ft}^2$$



5. Find the sum of the areas of the rectangle and triangle to find the area of the composite figure. Include units.

$$15 + 50 = 65 \text{ ft}^2$$

For each composite figure, find the area in square feet.

6. 	7. 
Triangle $\rightarrow 5 \cdot 7.5 \cdot 0.5 = 18.75$ Rectangle $\rightarrow 5 \cdot 12.5 = 62.5$ Total $\rightarrow 18.75 + 62.5 = 81.25$ Area = <u>81.25 ft²</u>	Triangle $\rightarrow \frac{1}{2} \cdot 5 \cdot 5 = 12.5 \text{ ft}^2$ Rectangle $\rightarrow 5 \cdot 3 = 15 \text{ ft}^2$ $12.5 + 15 = 27.5$ Area = <u>27.5 ft²</u>
8. 	9. 
Triangle₁ $\rightarrow 3 \cdot 2 \cdot \frac{1}{2} = 3$ $3 + 27 + 3 = 33$ Rectangle $\rightarrow 3 \cdot 9 = 27$ Triangle₂ $\rightarrow 3 \cdot 2 \cdot \frac{1}{2} = 3$ Area = <u>33 ft²</u>	Triangle $\rightarrow \frac{1}{2} \cdot 20 \cdot 5 = 50$ Rectangle $\rightarrow 10 \cdot 7.5 = 75$ $75 + 50 = 125$ Area = <u>125 ft²</u>

10. The local university is planting a garden. A drawing of the garden is shown. What is the area of the garden in square meters?

Rectangle₁ $\rightarrow 4 \cdot 8 = 32 \text{ m}^2$
 Rectangle₂ $\rightarrow 2 \cdot 5 = 10 \text{ m}^2$
 Rectangle₃ $\rightarrow 3 \cdot 5 = 15 \text{ m}^2$

$32 + 10 + 15 = 57 \text{ m}^2$

